



Matplotlib Plot Properties

A simple guide to the most useful `plt.plot()` and chart properties

Explained in easy, everyday English

Line Properties

These properties change how the line itself looks in a chart.

1 color

```
color='blue'
```

This changes the color of the line in your chart.

Examples:

```
color='red'  
color='green'  
color='black'  
color='orange'
```

2 linewidth

```
linewidth=3
```

This controls how thick or thin the line is.

Value	Meaning
1	Thin
3	Medium
6	Thick

3 linestyle

```
linestyle='--'
```

This controls the style or pattern of the line.

Style	Meaning
'-'	Solid line
'--'	Dashed line
'.'	Dotted line
'-.'	Dash-dot line

Example:

```
linestyle=':.'
```

4 marker

```
marker='o'
```

This shows a small symbol on every data point of the line.

Common markers:

Marker	Shape
'o'	Circle
'*'	Star
's'	Square
'^'	Triangle
'x'	Cross
'D'	Diamond

5 markersize

```
markersize=10
```

This changes how big or small the marker symbols are.

Value	Size
5	Small
10	Medium
20	Large

6 markerfacecolor

```
markerfacecolor='yellow'
```

This changes the inside (fill) color of the marker.

7 markeredgecolor

```
markeredgecolor='red'
```

This changes the color of the marker's border (outline).

8 markeredgewidth

```
markeredgewidth=2
```

This changes how thick the marker's border (outline) is.

9 label

```
label='Monthly Sales'
```

This gives a name to the line, which is used by `plt.legend()`. Without a label, the legend cannot show what the line means.

Chart & Axis Properties

These properties add titles, labels, and control what part of the chart is visible.

10 title

```
plt.title("Monthly Sales Report")
```

This adds a title to the top of the chart.

Useful extra properties:

```
fontsize=18  
fontweight='bold'  
color='blue'
```

11 xlabel()

```
plt.xlabel("Months")
```

This adds a label under the X-axis (the bottom line of the chart), so people know what it shows.

12 ylabel()

```
plt.ylabel("Sales")
```

This adds a label beside the Y-axis (the side line of the chart), so people know what it shows.

13 grid()

```
plt.grid(True)
```

This adds grid lines in the background of the chart, which makes it easier to read values.

Only horizontal lines:

```
plt.grid(axis='y')
```

Only vertical lines:

```
plt.grid(axis='x')
```

14 xlim()

```
plt.xlim('Jan', 'Jun')
```

This controls which part of the X-axis is visible on the chart.

For numeric data:

```
plt.xlim(0, 10)
```

15 ylim()

```
plt.ylim(100, 300)
```

This controls which part of the Y-axis is visible on the chart.

16 legend()

```
plt.legend()
```

This shows a small box with information about each line. It is very useful when your chart has more than one line.

17 annotate()

```
plt.annotate(  
    "Highest Sales",  
    xy=('Jun', 250),  
    xytext=('Apr', 270),  
    arrowprops=dict(facecolor='black')  
)
```

This is used to point out and highlight an important point on the chart, with text and an arrow. It is very common in reports and dashboards.

18 savefig()

```
plt.savefig("sales_report.png")
```

This saves the chart as an image file on your computer.

You can save it in different formats:

```
plt.savefig("sales.pdf")  
plt.savefig("sales.jpg")  
plt.savefig("sales.png")
```

19 show()

```
plt.show()
```

This displays the chart on your screen. It is usually the last line you write to actually see your plot.